

MASTER BUILD PROMPT — INSTITUTIONAL- GRADE MULTI-ASSET INVESTMENT AND ALGORITHMIC TRADING INTELLIGENCE PLATFORM

You are the principal architect, quantitative research lead, data engineering lead, machine learning lead, risk officer, execution engineer, security engineer, DevOps engineer, product designer, and technical program manager responsible for designing and implementing a private, institutional-grade multi-asset investment and algorithmic trading intelligence platform.

This platform will initially be used by one private operator. It is not currently a public SaaS product. However, its architecture must remain modular, auditable, secure, scalable, and capable of becoming a multi-user or institutional system later without requiring a complete rewrite.

Do not build a superficial trading dashboard, indicator collection, generic AI chatbot, or simple signal bot.

Build a professional quantitative research, investment intelligence, portfolio management, signal generation, paper-trading, and eventually controlled live-execution platform.

The platform must combine:

- Quantitative technical analysis
- Fundamental analysis
- Macroeconomic analysis
- Cross-asset analysis
- News and event intelligence
- Political and geopolitical event analysis
- Social-media sentiment and narrative analysis
- Market-regime detection
- Liquidity and market-microstructure analysis
- Strategy research and backtesting
- Portfolio construction
- Risk management
- Execution management
- Model validation
- Continuous monitoring
- Human-controlled AI assistance

The objective is not to maximize trade frequency, win rate, or short-term excitement.

The objective is to discover, validate, combine, and operate strategies that have a statistically defensible probability of generating positive risk-adjusted returns after realistic fees, spreads, slippage, funding costs, taxes where applicable, borrow costs, market impact, latency, and execution constraints.

No component may claim guaranteed profits.

The system must explicitly identify uncertainty, weak evidence, model risk, data gaps, regime dependency, execution risk, and the possibility of total or partial capital loss.

1. PRIMARY MISSION

Build a unified platform with two clearly separated but interconnected operating systems.

1.1 Active Trading System

The Active Trading System must research, test, rank, generate, monitor, and optionally execute trading strategies across multiple time horizons.

Supported horizons should include:

- Tick or event-driven, only when licensed and technically feasible
- 1-minute
- 5-minute
- 15-minute
- 30-minute
- 1-hour
- 4-hour
- Daily
- Weekly

The trading system must support:

- Intraday trading
- Swing trading
- Position trading
- Trend following
- Mean reversion
- Breakout trading
- Relative-value trading
- Statistical arbitrage
- Event-driven trading
- Macro trading
- Sentiment-driven strategies
- Volatility strategies
- Cross-asset confirmation
- Regime-dependent strategy selection

1.2 Investment and Portfolio System

Create a separate investment system for strategic and tactical portfolio management.

Supported investment horizons must include:

- 1 month

- 3 months
- 6 months
- 12 months
- 24 months
- 36 months
- 5 years or longer where sufficient data exists

The investment system must:

- Identify investable themes
- Rank industries, countries, asset classes, factors, and securities
- Analyze macroeconomic cycles
- Monitor valuation and fundamental quality
- Construct diversified portfolios
- Recommend rebalancing actions
- Compare current holdings against superior alternatives
- Measure portfolio concentration and hidden factor exposure
- Estimate downside and scenario risk
- Explain why an allocation should be maintained, increased, reduced, hedged, or removed
- Keep investment decisions separate from short-term trading noise

Trading capital and investment capital must use separate accounts, risk budgets, policies, position limits, performance reports, and decision processes.

2. NON-NEGOTIABLE OPERATING PRINCIPLES

Implement the following principles as hard requirements.

2.1 Capital Preservation First

The priority hierarchy is:

1. Prevent catastrophic loss.
2. Prevent uncontrolled execution.
3. Preserve data and audit integrity.
4. Maintain system availability and recoverability.
5. Produce robust risk-adjusted returns.
6. Improve absolute returns.
7. Increase strategy capacity and market coverage.

Never reverse this hierarchy.

2.2 Evidence Before Complexity

Begin with transparent baselines.

Do not use deep learning, reinforcement learning, autonomous agents, transformers, or complex ensemble systems merely because they appear sophisticated.

Every complex model must outperform appropriate simple baselines after:

- Out-of-sample testing
- Transaction costs
- Slippage
- Latency
- Multiple-testing correction
- Regime analysis
- Stability analysis
- Capacity analysis

2.3 Research and Production Separation

Maintain strict separation among:

- Raw data
- Cleaned data
- Point-in-time research data
- Training data
- Validation data
- Test data
- Paper-trading data
- Shadow-production data
- Live-production data

A production strategy must never be altered directly from an exploratory notebook.

2.4 No Look-Ahead or Survivorship Bias

Every dataset, feature, strategy, backtest, and model must be evaluated for:

- Look-ahead bias
- Survivorship bias
- Delisting bias
- Selection bias
- Publication-delay bias
- Revised macroeconomic data
- Corporate-action errors
- Future constituent leakage
- Time-zone leakage
- Incomplete market-calendar handling
- Duplicate or stale data
- Data snooping
- Overfitting

2.5 Human-Controlled Live Trading

The system may eventually execute orders, but live trading must remain disabled by default.

Live execution may only be enabled after:

- Paper trading is stable.
- Shadow-mode results are acceptable.
- Risk limits are configured.
- Broker reconciliation works.
- The kill switch has been tested.
- Failure simulations pass.
- The user explicitly enables live trading.
- The user explicitly approves the account and strategy.
- The user explicitly sets a maximum capital allocation.

Artificial intelligence may recommend actions but may not independently remove safety limits, increase risk budgets, enable leverage, connect new accounts, expose secrets, or activate live execution.

2.6 Full Auditability

Every material decision must be reproducible.

Store:

- Data versions
- Feature versions
- Model versions
- Strategy versions
- Configuration versions
- Prompt versions
- Signal inputs
- Signal outputs
- Risk decisions
- Order decisions
- Broker responses
- Fills
- Rejections
- Position changes
- Manual overrides
- User approvals
- Deployment versions

3. INITIAL REPOSITORY AND PLANNING INSTRUCTIONS

Before implementing features:

1. Inspect the entire existing repository.
2. Identify the current stack, modules, tests, deployment configuration, data models, secrets handling, and technical debt.

3. Do not delete or replace functioning code without documented justification.
4. Create a repository map.
5. Identify reusable components.
6. Identify security and data-integrity risks.
7. Identify architecture gaps.
8. Identify all external dependencies and data licenses.
9. Produce an implementation roadmap.
10. Begin implementation only after documenting the architecture.

Create the following files:

```
/docs/MASTER_ROADMAP.md
/docs/PRODUCT_REQUIREMENTS.md
/docs/SYSTEM_ARCHITECTURE.md
/docs/DATA_ARCHITECTURE.md
/docs/QUANT_RESEARCH_STANDARD.md
/docs/BACKTESTING_STANDARD.md
/docs/RISK_MANAGEMENT_POLICY.md
/docs/EXECUTION_POLICY.md
/docs/MODEL_GOVERNANCE.md
/docs/SECURITY_MODEL.md
/docs/OBSERVABILITY_PLAN.md
/docs/DATA_PROVIDER_MATRIX.md
/docs/BROKER_CAPABILITY_MATRIX.md
/docs/COMPLIANCE_CONSIDERATIONS.md
/docs/DISASTER_RECOVERY.md
/docs/TEST_STRATEGY.md
/docs/DEPLOYMENT_RUNBOOK.md
/docs/LIVE_TRADING_READINESS.md
/docs/DECISIONS.md
/docs/KNOWN_LIMITATIONS.md
/docs/CURRENT_STATUS.md
/docs/CHANGELOG.md
```

`MASTER_ROADMAP.md` must become the source of truth.

For every roadmap item include:

- Unique ID
- Description
- Business purpose
- Dependencies
- Risks
- Deliverables
- Tests
- Acceptance criteria
- Current status
- Evidence links
- Deferred work
- Estimated operational cost category

- Whether it blocks paper trading
- Whether it blocks live trading

Use status values:

```
NOT_STARTED
IN_PROGRESS
BLOCKED
IMPLEMENTED
TESTED
VERIFIED
DEFERRED
```

Never mark an item complete without objective evidence.

4. RECOMMENDED SYSTEM ARCHITECTURE

Use a modular event-driven architecture. Avoid unnecessary microservices in the first phase, but maintain clear service boundaries.

4.1 Core Technology Direction

Use the following default technical direction unless repository constraints justify a different choice.

Frontend

- Next.js
- TypeScript
- React
- Server-side authentication
- Responsive desktop-first interface
- Real-time WebSocket or server-sent event updates
- Professional financial charting library with verified commercial licensing
- Accessible design
- Strong information hierarchy
- Dark and light themes
- Keyboard-driven operator workflows

Application API

- Python
- FastAPI
- Strict request and response schemas
- Async support where useful
- OpenAPI documentation
- Role-based authorization
- Rate limiting
- Idempotency

- Structured error handling

Quantitative and ML Layer

- Python
- NumPy
- pandas or Polars
- SciPy
- scikit-learn
- statsmodels
- appropriate gradient-boosting libraries
- PyTorch only where justified
- vectorized and event-driven backtesting components
- reproducible experiment framework

Storage

Use purpose-specific storage:

- PostgreSQL for metadata, configuration, users, portfolios, orders, risk records, and audit data
- ClickHouse or another analytical time-series database for high-volume market and feature data
- Object storage for immutable Parquet datasets, model artifacts, reports, and backtest outputs
- Redis for caching, locks, rate limiting, and short-lived state
- Optional vector storage only for justified semantic retrieval use cases
- Secrets manager for credentials and signing keys

Messaging and Workflows

- Event streaming through Redpanda, Kafka, NATS, or an equivalent system
- Durable workflows through Temporal, Prefect, Dagster, or an equivalent orchestrator
- Scheduled jobs must be observable, retryable, idempotent, and recoverable
- Dead-letter queues must exist for unrecoverable events

Infrastructure

Provide:

- Local Docker Compose environment
- Development environment
- Test environment
- Paper-trading environment
- Production environment
- Infrastructure-as-code
- CI/CD
- Secret isolation
- Automated migrations
- Rollback procedures
- Backup and restore automation

Do not assume a single cloud provider until a cost, latency, compliance, reliability, and operational-complexity comparison has been documented.

5. DOMAIN-DRIVEN SERVICE BOUNDARIES

Design the system around the following logical services.

5.1 Instrument Master Service

Maintain canonical records for:

- Symbols
- Global identifiers
- Exchanges
- Venues
- Asset classes
- Quote currencies
- Base currencies
- Contract sizes
- Tick sizes
- Lot sizes
- Trading hours
- Holidays
- Time zones
- Corporate-action history
- Futures expirations
- Continuous futures mappings
- Crypto exchange pairs
- Forex pair conventions
- Delisted securities
- Symbol changes
- Broker-specific identifiers
- Data-provider-specific identifiers

A symbol string alone must never be treated as a globally unique instrument identifier.

5.2 Market Data Service

Ingest and normalize:

- Trades
- Quotes
- OHLCV bars
- Order-book snapshots
- Order-book deltas
- Auction data where available
- Indicative prices
- Open interest
- Funding rates
- Liquidation data
- Futures basis
- Options chains in a later phase
- Implied volatility in a later phase

- Reference rates
- Index values
- Market breadth
- ETF flows where legally accessible
- Short-interest data where legally accessible

5.3 Fundamental Data Service

Support point-in-time:

- Income statements
- Balance sheets
- Cash-flow statements
- Earnings releases
- Earnings estimates
- Revenue estimates
- Estimate revisions
- Guidance
- Valuation ratios
- Profitability
- Margins
- Growth
- Leverage
- Liquidity
- Capital allocation
- Share dilution
- Buybacks
- Dividends
- Corporate actions
- Insider transactions where legally available
- Institutional ownership where legally available
- Sector and industry classifications
- Country and currency exposure

Store both:

- As-reported values
- Standardized values

Preserve filing dates, effective dates, reporting periods, revision dates, and source provenance.

5.4 Macroeconomic Data Service

Ingest and version:

- Policy rates
- Central-bank decisions
- Central-bank statements
- Inflation
- Employment
- Unemployment

- Wage growth
- GDP
- PMIs
- Industrial production
- Retail sales
- Consumer confidence
- Producer prices
- Housing indicators
- Government debt
- Fiscal balances
- Yield curves
- Real yields
- Inflation expectations
- Credit spreads
- Money supply
- Liquidity measures
- Currency reserves
- Commodity inventories
- Trade balances
- Economic-surprise indicators
- Financial-conditions indicators

The service must distinguish:

- Observation period
- Release timestamp
- Expected value
- Initial reported value
- Revised value
- Prior value
- Consensus dispersion

Backtests must use only the information available at the historical decision timestamp.

5.5 News and Event Intelligence Service

Ingest legally accessible news and public events.

Capabilities:

- Duplicate detection
- Story clustering
- Entity recognition
- Entity linking
- Ticker linking
- Country linking
- Sector linking
- Event classification
- Source credibility scoring
- Publication timestamp validation
- Event timestamp extraction

- Novelty scoring
- Relevance scoring
- Sentiment scoring
- Uncertainty scoring
- Urgency scoring
- Expected impact horizon
- Contradiction detection
- Rumor labeling
- Follow-up tracking

Event categories must include:

- Earnings
- Guidance
- Mergers and acquisitions
- Regulatory actions
- Lawsuits
- Product launches
- Product failures
- Cybersecurity incidents
- Management changes
- Capital raises
- Buybacks
- Dividends
- Credit events
- Sovereign events
- Elections
- Sanctions
- Wars and conflicts
- Trade restrictions
- Central-bank events
- Natural disasters
- Supply disruptions
- Strikes
- Commodity shocks
- Exchange outages
- Protocol exploits
- Token unlocks
- Crypto governance changes
- ETF or regulatory decisions
- Fraud allegations

Never treat an unverified social-media post as confirmed news.

5.6 Social and Narrative Intelligence Service

Analyze only legally and contractually permitted sources.

Potential source categories:

- X

- Reddit
- Stock-focused communities
- Crypto communities
- Public Telegram channels where access is lawful
- Public YouTube metadata and transcripts where permitted
- Search-interest data
- News comments where permitted
- Public forums
- Public earnings-call transcripts

Calculate:

- Mention volume
- Mention acceleration
- Unique-author growth
- Bot-likelihood indicators
- Sentiment
- Sentiment change
- Disagreement
- Emotional intensity
- Narrative clustering
- Emerging-topic detection
- Influencer concentration
- Geographic distribution where lawful
- Language distribution
- Spam concentration
- Coordinated-promotion likelihood
- Pump-and-dump risk indicators
- Narrative persistence
- Price-versus-sentiment divergence

Separate:

- Retail sentiment
- Professional commentary
- News amplification
- Automated spam
- Promotional activity
- Organic discussion

Do not use social sentiment as a standalone trade trigger.

5.7 Feature Platform

Create a versioned feature platform containing at least the following feature families.

Price and Return Features

- Arithmetic returns
- Log returns
- Multi-horizon returns

- Overnight returns
- Intraday returns
- Gap behavior
- Relative returns
- Residual returns
- Cross-sectional rank
- Trend strength
- Acceleration
- Distance from highs and lows

Momentum Features

- Time-series momentum
- Cross-sectional momentum
- Relative strength
- Moving-average relationships
- Breakout persistence
- Trend consistency
- Momentum quality
- Momentum crashes
- Residual momentum

Mean-Reversion Features

- Z-scores
- Deviation from moving averages
- Deviation from VWAP
- Bollinger-type deviations
- Residual spread
- Cointegration residual
- Short-horizon reversal
- Overnight reversal
- Intraday reversal
- Liquidity-adjusted reversal

Volatility Features

- Realized volatility
- Parkinson volatility
- Garman–Klass volatility
- Range measures
- Volatility-of-volatility
- Downside volatility
- Upside volatility
- Jump detection
- Volatility term structure
- Implied-versus-realized volatility where available

Volume and Liquidity Features

- Relative volume
- Volume trend

- Dollar volume
- Turnover
- Bid-ask spread
- Effective spread
- Quote depth
- Order-book imbalance
- Trade imbalance
- Amihud-type illiquidity
- Price impact
- Volume profile
- VWAP deviation
- Participation rates
- Liquidity regime

Market Breadth Features

- Advance-decline measures
- Percentage above moving averages
- New highs versus new lows
- Equal-weight versus capitalization-weight divergence
- Sector participation
- Cross-sectional dispersion
- Correlation breadth
- Market concentration

Derivatives and Crypto Features

- Funding rate
- Funding-rate change
- Open interest
- Open-interest change
- Liquidations
- Futures basis
- Perpetual basis
- Spot-perpetual divergence
- Exchange flow
- Stablecoin flow where reliable
- Token unlock schedule
- Staking yield
- Borrow cost
- Options skew and term structure in a later phase

Fundamental Features

- Valuation
- Growth
- Quality
- Profitability
- Leverage
- Liquidity
- Accruals
- Cash conversion

- Earnings quality
- Estimate revisions
- Earnings surprise
- Guidance surprise
- Capital efficiency
- Shareholder yield
- Dilution
- Financial distress
- Balance-sheet momentum

Macro Features

- Growth regime
- Inflation regime
- Policy regime
- Liquidity regime
- Yield-curve level, slope, and curvature
- Real-rate direction
- Credit-spread direction
- Dollar regime
- Commodity regime
- Economic-surprise momentum
- Financial-conditions momentum

Sentiment and Event Features

- News sentiment
- Social sentiment
- Sentiment change
- Mention acceleration
- Event novelty
- Source confidence
- Entity relevance
- Event severity
- Narrative persistence
- Sentiment-price divergence
- News-volume shock

Cross-Asset Features

- Equity-rate relationship
- Equity-credit relationship
- Gold-real-yield relationship
- Oil-inflation relationship
- Crypto-liquidity relationship
- Currency-rate-differential relationship
- Commodity-currency relationships
- Risk-on/risk-off composites
- Correlation breaks
- Lead-lag relationships

Each feature must have:

- Definition
 - Owner
 - Source
 - Frequency
 - Timestamp semantics
 - Lookback
 - Missing-value policy
 - Outlier policy
 - Version
 - Test coverage
 - Leakage test
 - Expected range
 - Data-quality metrics
-

6. ASSET-CLASS COVERAGE

Design independent adapters for each market.

6.1 United States Equities

Support:

- Listed equities
- ETFs
- Major indices
- Sector and industry universes
- Corporate actions
- Earnings events
- Extended-hours data where available
- Delisted-symbol history
- Survivorship-free universes

6.2 Turkish Equities

Support:

- Borsa İstanbul instruments
- TRY currency effects
- Turkish market calendars
- Price limits
- Local corporate disclosures
- Capital increases
- Dividend adjustments
- Inflation-aware analysis
- Currency-adjusted performance
- Local liquidity constraints

Do not assume United States market rules apply to Borsa İstanbul.

6.3 Forex

Support:

- Major pairs
- Minor pairs
- Selected emerging-market pairs
- Interest-rate differentials
- Carry
- Rollover costs
- Session effects
- Central-bank events
- Macro-surprise analysis
- Broker spread variation
- Quote-convention normalization

6.4 Precious Metals and Commodities

Support:

- Gold
- Silver
- Oil
- Natural gas
- Selected industrial metals
- Selected agricultural products where data is available

Distinguish:

- Spot
- Futures
- ETFs
- CFDs
- Physically backed products

Model:

- Contract expiration
- Roll yield
- Contango
- Backwardation
- Storage effects
- Seasonal behavior

6.5 Cryptocurrencies

Support multiple exchanges through normalized adapters.

Include:

- Spot
- Perpetual futures
- Dated futures
- Funding
- Open interest
- Liquidations
- Basis
- Exchange fragmentation
- Stablecoin pairs
- Token metadata
- Token unlocks
- Market-cap history
- Supply changes
- Exchange outages
- Delistings
- Protocol incidents

Do not treat all exchanges as equivalent.

Model:

- Venue risk
- Counterparty risk
- Fragmented liquidity
- 24/7 market calendars
- Exchange-specific prices
- Transfer and withdrawal risk
- Stablecoin risk
- Extreme gap and liquidation behavior

6.6 Optional Later Asset Classes

Architect but do not initially prioritize:

- Options
- Fixed income
- Futures spreads
- Volatility products
- European equities
- Asian equities
- Prediction markets

Each optional market requires a separate research, data, legal, risk, and execution review before activation.

7. DATA-PROVIDER AND BROKER EVALUATION

Do not select providers by popularity alone.

Create `/docs/DATA_PROVIDER_MATRIX.md`.

For every provider evaluate:

- Covered markets
- Covered instruments
- Historical depth
- Real-time availability
- Tick availability
- Quote availability
- Order-book availability
- Fundamental coverage
- News coverage
- Macro coverage
- Corporate actions
- Delistings
- Point-in-time correctness
- Survivorship-free support
- Revision history
- API limits
- WebSocket support
- Expected latency
- Service-level commitments
- Licensing rights
- Storage rights
- Redistribution restrictions
- Commercial-use restrictions
- Geographic restrictions
- Monthly cost
- Overage cost
- Reliability
- Support quality
- Vendor lock-in
- Backup-provider availability

Create `/docs/BROKER_CAPABILITY_MATRIX.md`.

For every broker or exchange evaluate:

- Supported asset classes
- Countries accepted
- Account requirements
- Paper trading
- Order types
- Fractional trading
- Short selling

- Margin
- Leverage
- Market-data requirements
- API authentication
- API stability
- Rate limits
- Streaming
- Order-status events
- Fill events
- Account events
- Position reconciliation
- Corporate-action handling
- Extended-hours support
- Cancel/replace behavior
- Idempotency support
- Native bracket orders
- Stop orders
- Trailing stops
- OCO orders
- Availability in Belgium
- Tax-reporting implications
- Currency conversion
- Withdrawal controls

Build broker adapters behind one internal execution interface.

Never hardcode business logic to one broker.

8. DATA QUALITY AND PROVENANCE

Create automated checks for:

- Missing bars
- Duplicate bars
- Negative prices
- Impossible volumes
- Crossed quotes
- Stale quotes
- Time regressions
- Time-zone mismatches
- Corporate-action discontinuities
- Sudden unexplained symbol changes
- Price divergence across providers
- Inconsistent currency units
- Delayed macro timestamps
- Revised data
- Abnormal spread values
- Incomplete order books
- Missing funding rates

- Exchange outages
- News duplicates
- Social spam bursts

Each data record must contain sufficient provenance:

- Provider
- Source identifier
- Ingestion timestamp
- Event timestamp
- Effective timestamp
- Original timezone
- Normalized timestamp
- Revision number
- Data version
- Quality score
- Processing status

Generate a Data Health Score for every instrument and dataset.

Strategies must not trade from data below their configured quality threshold.

9. STRATEGY RESEARCH LIBRARY

Create a modular strategy framework.

Every strategy must implement standardized interfaces for:

- Universe selection
- Required datasets
- Feature generation
- Signal generation
- Position sizing
- Entry logic
- Exit logic
- Stop logic
- Holding period
- Risk limits
- Order generation
- Cost model
- Capacity model
- Benchmark
- Explanation
- Versioning

Research the following strategy families.

9.1 Trend-Following Strategies

- Moving-average trend
- Breakout trend
- Donchian-style channels
- Time-series momentum
- Volatility-adjusted trend
- Multi-timeframe trend
- Trend with breadth confirmation
- Trend with macro confirmation
- Trend with sentiment confirmation
- Trend with volatility filters

9.2 Mean-Reversion Strategies

- Short-term reversal
- VWAP reversion
- Range reversion
- Overnight gap reversion
- Volatility-normalized reversion
- Index constituent reversion
- Residual reversion
- Liquidity-shock reversion
- Sentiment-exhaustion reversion

9.3 Relative-Value and Statistical-Arbitrage Strategies

- Pairs trading
- Cointegration
- Sector-neutral spreads
- Market-neutral residual strategies
- Cross-exchange crypto spreads
- Spot-futures basis
- Perpetual-funding strategies
- ETF versus constituents
- Dual-listed instruments where executable
- Cross-sectional factor spreads

Require explicit borrow, execution, timing, and capacity models.

9.4 Breakout and Volatility-Expansion Strategies

- Price breakouts
- Volume-confirmed breakouts
- Volatility compression and expansion
- Opening-range breakouts
- Event-driven breakouts
- Order-book imbalance breakouts
- Failed-breakout reversal

9.5 Fundamental Strategies

- Value
- Quality
- Growth
- Profitability
- Earnings revisions
- Earnings momentum
- Shareholder yield
- Low investment
- Financial strength
- Composite factor models

9.6 Macro Strategies

- Growth and inflation regime allocation
- Yield-curve strategies
- Real-rate-sensitive allocation
- Currency carry
- Commodity macro
- Risk-on/risk-off allocation
- Liquidity-cycle allocation
- Economic-surprise strategies
- Central-bank divergence strategies

9.7 Event-Driven Strategies

- Earnings surprise
- Guidance change
- Analyst-revision momentum
- Merger events
- Regulatory decisions
- Index inclusions and exclusions
- Buybacks
- Capital raises
- Token unlocks
- Exchange listings and delistings
- Protocol incidents
- Central-bank decisions
- Election and geopolitical events

9.8 Sentiment and Narrative Strategies

- Sentiment momentum
- Sentiment reversal
- Mention-volume shock
- News novelty
- Narrative acceleration
- Price-sentiment divergence
- Retail-crowding reversal

- Confirmed-event sentiment continuation

Sentiment strategies require manipulation and spam filters.

9.9 Cross-Asset Strategies

- Equity and bond confirmation
- Gold and real-yield relationships
- Oil and inflation relationships
- Credit and equity divergence
- Dollar and commodities
- Crypto and global-liquidity relationships
- Currency and rate differentials
- Sector rotation

9.10 Defensive and Crisis Strategies

- Volatility targeting
- Trend-based de-risking
- Drawdown control
- Tail-risk proxies
- Cash allocation
- Gold allocation
- Duration allocation
- Correlation-break detection
- Liquidity-crisis detection

Do not assume one strategy works permanently.

10. MARKET-REGIME ENGINE

Create a dedicated regime engine.

It must classify markets across multiple dimensions.

10.1 Regime Dimensions

- Bull trend
- Bear trend
- Sideways market
- High volatility
- Low volatility
- Volatility expansion
- Volatility contraction
- High liquidity
- Low liquidity
- Risk-on
- Risk-off
- Inflationary growth

- Inflationary contraction
- Disinflationary growth
- Deflationary contraction
- Tightening liquidity
- Easing liquidity
- Credit stress
- Currency stress
- Commodity shock
- Correlation breakdown
- Crisis regime

10.2 Regime Methods

Compare:

- Rule-based classification
- Hidden Markov models
- Change-point detection
- Clustering
- Bayesian state models
- Tree-based classification
- Probabilistic ensembles

The regime output must include probabilities, not only hard labels.

10.3 Regime-Dependent Strategy Allocation

The system may:

- Enable a strategy
- Disable a strategy
- Reduce strategy risk
- Increase strategy risk within pre-approved bounds
- Change holding period
- Change stop logic
- Change universe
- Change execution style

Regime decisions must be validated out of sample.

Do not create a regime engine that merely explains past markets well but fails to identify regimes in real time.

11. BACKTESTING ENGINE

Build both vectorized and event-driven backtesting capabilities.

11.1 Required Simulation Features

Support:

- Market orders
- Limit orders
- Stop orders
- Stop-limit orders
- Bracket orders
- Partial fills
- Rejections
- Cancellations
- Cancel/replace
- Expired orders
- Session boundaries
- Trading halts
- Price limits
- Corporate actions
- Dividends
- Splits
- Symbol changes
- Delistings
- Futures expiration
- Futures rolls
- Funding payments
- Borrow costs
- Margin
- Interest on cash
- Currency conversion
- Exchange fees
- Broker fees
- Taxes where applicable
- Bid-ask spread
- Slippage
- Market impact
- Latency
- Queue assumptions where possible

11.2 Cost Models

Provide at least:

- Fixed cost
- Percentage cost
- Spread-based cost
- Volatility-based slippage
- Volume participation model
- Square-root market-impact approximation
- Instrument-specific calibrated model

Run pessimistic, base, and optimistic cost scenarios.

The production decision must use the base and pessimistic cases, not only optimistic assumptions.

11.3 Validation Framework

Use:

- Chronological train-validation-test splits
- Rolling walk-forward analysis
- Expanding-window analysis
- Purged cross-validation
- Embargo periods
- Nested validation where model selection occurs
- Parameter-stability analysis
- Bootstrapping
- Monte Carlo trade-sequence analysis
- Regime-specific validation
- Instrument-level validation
- Cross-market validation
- Crisis-period stress tests
- Delayed-signal tests
- Increased-cost tests
- Missing-data tests

11.4 Multiple-Testing Protection

Track:

- Number of strategies tested
- Number of parameter combinations
- Number of universes tested
- Number of feature combinations
- Number of rejected experiments
- Researcher degrees of freedom

Use appropriate methods such as:

- Deflated Sharpe Ratio
- Probabilistic Sharpe Ratio
- False-discovery controls
- Reality-check methods
- Probability of backtest overfitting
- Holdout periods not used during development

Never select a strategy only because it has the best historical result among thousands of trials.

11.5 Required Performance Metrics

Calculate:

- Total return

- Annualized return
- CAGR
- Annualized volatility
- Sharpe ratio
- Sortino ratio
- Calmar ratio
- Information ratio
- Omega ratio where useful
- Maximum drawdown
- Average drawdown
- Drawdown duration
- Time under water
- Ulcer index
- Value at Risk
- Conditional Value at Risk
- Tail ratio
- Skew
- Kurtosis
- Hit rate
- Payoff ratio
- Profit factor
- Expectancy
- Average win
- Average loss
- Largest win
- Largest loss
- Consecutive losses
- Consecutive wins
- Exposure
- Turnover
- Capacity
- Slippage sensitivity
- Cost sensitivity
- Beta
- Alpha
- Upside capture
- Downside capture
- Monthly win rate
- Rolling performance
- Regime performance

Never rank strategies by win rate alone.

A strategy with a lower win rate may be superior if its expectancy, payoff ratio, drawdown, tail behavior, and robustness are stronger.

12. STRATEGY SCORECARD

Create a standardized Strategy Quality Score.

Possible components:

- Out-of-sample return
- Out-of-sample Sharpe
- Drawdown
- Return-to-drawdown
- Statistical confidence
- Parameter stability
- Cross-market stability
- Regime diversity
- Cost robustness
- Slippage robustness
- Capacity
- Turnover
- Signal decay
- Tail risk
- Correlation with current portfolio
- Live versus backtest consistency
- Data quality
- Model simplicity
- Explainability

Apply penalties for:

- Excessive complexity
- Narrow parameter peaks
- High turnover
- Small sample sizes
- Weak out-of-sample results
- High dependence on one event
- High dependence on one instrument
- Unrealistic fills
- Illiquidity
- Strong performance only before costs
- Excessive leverage
- Negative recent drift
- Unstable feature importance

Display score components separately. Do not hide them behind one number.

13. MACHINE LEARNING AND AI POLICY

Machine learning must support the research process, not replace scientific validation.

13.1 Model Hierarchy

Test in this order:

1. Naive baseline

2. Rule-based baseline
3. Linear or logistic model
4. Regularized linear model
5. Tree-based model
6. Gradient boosting
7. Time-series model
8. Neural model
9. Ensemble
10. Reinforcement learning only under strict justification

Do not move to the next complexity level unless measurable value has been demonstrated.

13.2 Supported ML Tasks

- Return-direction probability
- Expected-return estimation
- Volatility forecasting
- Drawdown-risk forecasting
- Regime classification
- Event-impact classification
- Sentiment classification
- Anomaly detection
- Liquidity forecasting
- Slippage forecasting
- Strategy-selection assistance
- Position-sizing assistance
- Portfolio-risk forecasting

13.3 Prediction Output

Every model prediction must include:

- Timestamp
- Instrument
- Horizon
- Prediction
- Probability or confidence
- Calibration information
- Model version
- Feature version
- Data version
- Regime
- Explanation
- Expected uncertainty
- Expiration time

13.4 Model Validation

Evaluate:

- Calibration

- Precision
- Recall
- F1 where applicable
- ROC-AUC where appropriate
- PR-AUC for imbalanced problems
- Brier score
- Log loss
- Rank correlation
- Information coefficient
- Stability
- Drift
- Economic value after costs

A statistically accurate model may still be economically useless.

13.5 Explainability

Provide:

- Feature importance
- Permutation importance
- SHAP-type explanations where appropriate
- Local prediction explanation
- Historical analogues
- Scenario sensitivity
- Confidence degradation

Do not present post-hoc explanations as causal proof.

13.6 Drift Monitoring

Monitor:

- Data drift
- Feature drift
- Prediction drift
- Calibration drift
- Performance drift
- Regime drift
- Execution drift
- Cost drift

A model may be automatically reduced or disabled when drift thresholds are exceeded, but automatic risk increases are prohibited.

13.7 Large Language Model Usage

LLMs may be used for:

- News summarization
- Event extraction

- Entity linking
- Report generation
- Strategy explanation
- Research assistance
- Natural-language querying
- Log investigation
- Documentation
- Code review assistance

LLMs must not be the sole authority for:

- Order placement
- Position sizing
- Legal interpretation
- Source verification
- Risk-limit changes
- Live-trading activation

LLM outputs must be treated as untrusted structured proposals and validated through deterministic rules.

14. SIGNAL ENGINE

Every signal must include:

- Instrument
- Asset class
- Strategy
- Direction
- Entry zone
- Invalidation level
- Stop level
- Target logic
- Expected holding period
- Signal strength
- Confidence
- Expected return
- Expected volatility
- Expected downside
- Risk-reward estimate
- Regime
- Liquidity score
- Data quality score
- News risk
- Event risk
- Correlation impact
- Position-size recommendation
- Expiration time
- Explanation

- Contradicting evidence

Signal statuses:

```
CANDIDATE
VALIDATED
BLOCKED_BY_RISK
BLOCKED_BY_DATA
WAITING_FOR_ENTRY
ACTIVE
PARTIALLY_FILLED
FILLED
INVALIDATED
EXPIRED
CLOSED
CANCELLED
```

A signal is not automatically an order.

The signal must pass:

1. Data validation
2. Strategy validation
3. Regime validation
4. Liquidity validation
5. Portfolio validation
6. Risk validation
7. Broker validation
8. Execution validation
9. User-policy validation

15. STRATEGY ENSEMBLE AND CAPITAL ALLOCATION

Create an ensemble layer that combines strategies without allowing one unstable model to dominate.

Potential approaches:

- Equal weighting
- Volatility weighting
- Inverse-risk weighting
- Risk parity
- Hierarchical risk parity
- Constrained mean-variance
- Drawdown-aware allocation
- Bayesian model averaging
- Performance-weighted allocation

- Correlation-aware strategy allocation
- Regime-conditioned allocation

Allocation constraints must include:

- Maximum strategy weight
- Maximum asset-class weight
- Maximum instrument weight
- Maximum sector weight
- Maximum country weight
- Maximum currency exposure
- Maximum broker exposure
- Maximum exchange exposure
- Maximum correlated-cluster exposure
- Maximum leverage
- Maximum turnover
- Minimum liquidity
- Maximum expected shortfall

Recent performance may influence weights, but short-term performance chasing must be prevented.

Use decay, confidence intervals, minimum sample sizes, and turnover penalties.

16. RISK MANAGEMENT ENGINE

The Risk Engine must be independent from the Strategy Engine.

The Strategy Engine proposes.

The Risk Engine approves, modifies, reduces, delays, or rejects.

16.1 Position-Level Controls

- Maximum position notional
- Maximum percentage of equity
- Maximum volatility contribution
- Maximum loss per trade
- Maximum gap risk
- Stop-loss policy
- Time stop
- Liquidity threshold
- Spread threshold
- Event-risk threshold
- Concentration threshold

16.2 Portfolio-Level Controls

- Gross exposure
- Net exposure

- Leverage
- Asset-class exposure
- Sector exposure
- Country exposure
- Currency exposure
- Factor exposure
- Correlation concentration
- Beta
- Duration
- Commodity exposure
- Crypto exposure
- Stablecoin exposure
- Broker exposure
- Exchange exposure
- Counterparty exposure
- VaR
- CVaR
- Drawdown
- Stress loss

16.3 Daily and Periodic Limits

- Daily loss limit
- Weekly loss limit
- Monthly loss limit
- Strategy loss limit
- Broker loss limit
- Asset-class loss limit
- Consecutive-loss limit
- Abnormal-slippage limit
- Order-rejection limit
- Data-error limit
- System-error limit

16.4 Kill Switches

Implement:

- Global kill switch
- Broker kill switch
- Exchange kill switch
- Strategy kill switch
- Instrument kill switch
- Asset-class kill switch
- Data-provider kill switch
- Model-version kill switch

Kill switches must:

- Prevent new orders
- Cancel eligible working orders

- Preserve logs
- Notify the operator
- Require explicit reactivation
- Never be disabled automatically

16.5 Pre-Trade Risk Controls

Reject or hold orders when:

- Price is stale.
- Quote is missing.
- Spread is excessive.
- Order size exceeds limits.
- Position limit would be breached.
- Daily loss limit is reached.
- Instrument is halted.
- Trading session is invalid.
- Data quality is insufficient.
- Broker state is uncertain.
- Account reconciliation is incomplete.
- Duplicate-order risk exists.
- Strategy is disabled.
- Model version is not approved.
- Signal has expired.
- News or event risk exceeds limits.
- Expected slippage is excessive.
- Available buying power is insufficient.
- Currency conversion is unresolved.

16.6 Stress Tests

Run scenarios including:

- 1987-type equity crash
- 2008-type credit crisis
- 2020-type liquidity shock
- Rapid rate increases
- Currency devaluation
- Gold shock
- Oil shock
- Crypto exchange failure
- Stablecoin depeg
- Flash crash
- Exchange outage
- Broker outage
- Data-provider outage
- Internet loss
- Database delay
- Duplicated events
- Delayed fills
- Rejected orders

- Partial fills
- Sudden correlation convergence
- Extreme spread expansion

Historical labels are examples. Implement generalized scenario definitions rather than hardcoded narratives only.

17. EXECUTION MANAGEMENT SYSTEM

Create a broker-neutral Order Management System and Execution Management System.

17.1 Order Lifecycle

Support:

```
PROPOSED
RISK_PENDING
RISK_REJECTED
APPROVED
SUBMISSION_PENDING
SUBMITTED
ACKNOWLEDGED
PARTIALLY_FILLED
FILLED
CANCEL_PENDING
CANCELLED
REJECTED
EXPIRED
UNKNOWN
RECONCILIATION_REQUIRED
```

17.2 Execution Algorithms

Implement only after basic order reliability is complete:

- Market
- Passive limit
- Aggressive limit
- TWAP
- VWAP
- Percentage-of-volume
- Iceberg where supported
- Adaptive participation
- Spread-aware execution
- Urgency-based execution

17.3 Execution Quality

Measure:

- Arrival-price slippage
- VWAP slippage
- Decision-price slippage
- Implementation shortfall
- Fill ratio
- Time to fill
- Rejection rate
- Cancel rate
- Partial-fill rate
- Spread paid
- Market impact
- Opportunity cost
- Broker comparison

17.4 Idempotency and Duplicate Protection

Every order intent must have an immutable internal identifier.

Before submitting an order:

- Check existing intent.
- Check active broker orders.
- Check recent fills.
- Check current position.
- Check account state.
- Check reconciliation state.

Network retries must never create duplicate exposure.

17.5 Reconciliation

Continuously compare:

- Internal cash versus broker cash
- Internal positions versus broker positions
- Internal open orders versus broker orders
- Internal fills versus broker fills
- Internal realized P&L versus broker P&L
- Internal fees versus broker fees

Any unresolved difference must block risk increases.

18. PAPER TRADING AND LIVE-READINESS STAGES

Use the following promotion stages.

Stage 0 — Research Only

- Historical data
- Feature calculation
- Backtests
- No broker connection

Stage 1 — Simulated Paper Trading

- Internal fill simulator
- Live market data
- No broker orders

Stage 2 — Broker Paper Trading

- Broker sandbox or paper account
- Real order lifecycle
- No real capital

Stage 3 — Shadow Mode

- Generate live orders
- Do not submit them
- Compare hypothetical fills with real markets

Stage 4 — Limited Live Pilot

- One approved strategy
- One approved asset class
- Minimal capital
- No leverage
- Strict limits
- Manual approval optional or mandatory

Stage 5 — Controlled Live Multi-Strategy

- Multiple verified strategies
- Independent risk budgets
- Automated risk controls
- Continuous reconciliation

No stage may be skipped.

Create objective promotion and rollback criteria.

19. INVESTMENT INTELLIGENCE MODULE

Build a professional long-term investment workspace.

19.1 Theme Discovery

Track themes such as:

- Artificial intelligence
- Robotics
- Automation
- Semiconductors
- Energy transition
- Nuclear energy
- Grid modernization
- Cybersecurity
- Space
- Defense
- Biotechnology
- Longevity
- Tokenization
- Digital payments
- Cloud infrastructure
- Data centers
- Critical minerals
- Water
- Emerging markets
- Demographic changes

Themes are examples, not permanent recommendations.

For each theme calculate:

- Economic relevance
- Revenue exposure
- Earnings exposure
- Capital expenditure trend
- Policy support
- Regulatory risk
- Competitive structure
- Valuation
- Investor crowding
- Momentum
- Sentiment
- Adoption curve
- Supply constraints
- Geographic exposure
- Currency exposure
- Theme purity
- Theme maturity

- Expected horizon
- Primary risks

19.2 Company and Asset Scoring

Score:

- Quality
- Growth
- Valuation
- Profitability
- Balance-sheet strength
- Cash-flow quality
- Competitive position
- Capital allocation
- Earnings revisions
- Price momentum
- Risk
- Liquidity
- Governance
- Theme exposure
- Macro sensitivity

Do not compress all analysis into a single unexplained score.

19.3 Portfolio Construction

Support:

- Strategic asset allocation
- Tactical asset allocation
- Core-satellite portfolios
- Factor portfolios
- Theme portfolios
- Risk-parity portfolios
- Maximum-diversification portfolios
- Drawdown-aware portfolios
- Currency-hedged and unhedged alternatives
- Benchmark-relative portfolios

19.4 Rebalancing

Evaluate monthly, but only trade when benefits exceed costs.

Possible triggers:

- Calendar rebalance
- Allocation drift
- Risk drift
- Thesis deterioration

- Valuation change
- Fundamental change
- Regime change
- Correlation change
- Better replacement candidate
- Drawdown threshold
- Concentration threshold

Every recommendation must show:

- Current allocation
- Proposed allocation
- Reason
- Expected benefit
- Expected cost
- Tax considerations where data exists
- Risk impact
- Turnover
- Confidence
- Alternative action

19.5 Investment Thesis Records

For every holding store:

- Thesis
- Entry date
- Expected horizon
- Expected catalysts
- Valuation framework
- Fundamental assumptions
- Primary risks
- Invalidation criteria
- Review schedule
- Position-size rationale
- Exit criteria
- Relevant documents
- Decision history

20. PROFESSIONAL OPERATOR DASHBOARD

Create a highly functional private command center.

Do not prioritize decorative animations over speed, clarity, stability, or decision quality.

20.1 Main Command Center

Display:

- Total portfolio value
- Trading capital
- Investment capital
- Daily P&L
- Realized P&L
- Unrealized P&L
- Current drawdown
- Gross exposure
- Net exposure
- Leverage
- VaR
- CVaR
- Active risk alerts
- Data-health alerts
- Broker status
- Strategy status
- Market-regime summary
- Highest-priority signals
- Important upcoming events
- Current positions
- Open orders
- Recent fills
- System-health summary

20.2 Market Intelligence Page

Include:

- Global market heatmap
- Asset-class performance
- Sector performance
- Country performance
- Rates
- Yield curves
- Credit
- Currencies
- Commodities
- Crypto
- Volatility
- Breadth
- Correlations
- Liquidity indicators
- Macro regime
- Risk-on/risk-off score

20.3 Instrument Workstation

For each instrument show:

- Multi-timeframe chart
- Price and volume
- Indicators
- Order book where available
- Volume profile
- Fundamentals
- Valuation
- News
- Social sentiment
- Event timeline
- Related assets
- Correlations
- Macro sensitivities
- Active signals
- Strategy history
- Backtest summaries
- Current position
- Orders
- Risk
- AI-generated explanation with source references

20.4 Signal Explorer

Filters:

- Asset class
- Market
- Strategy
- Direction
- Horizon
- Confidence
- Regime
- Liquidity
- Expected return
- Expected risk
- Risk-reward
- Data quality
- Event risk
- Status

20.5 Strategy Laboratory

Allow:

- Strategy configuration
- Universe definition

- Parameter ranges
- Feature selection
- Cost assumptions
- Risk settings
- Backtest launch
- Walk-forward testing
- Stress testing
- Comparison
- Experiment tracking
- Promotion requests

20.6 Backtest Analysis Page

Show:

- Equity curve
- Drawdown curve
- Rolling Sharpe
- Rolling volatility
- Monthly returns
- Return distribution
- Trade distribution
- Regime performance
- Instrument performance
- Cost sensitivity
- Parameter heatmaps
- Walk-forward segments
- In-sample versus out-of-sample
- Monte Carlo outcomes
- Benchmark comparison

20.7 Risk Center

Show:

- Exposure map
- Correlation clusters
- Factor exposures
- Scenario losses
- VaR and CVaR
- Drawdown
- Limit usage
- Concentration
- Liquidity risk
- Counterparty risk
- Broker risk
- Exchange risk
- Current blocks
- Kill-switch controls

20.8 Portfolio and Investment Center

Show:

- Strategic allocation
- Tactical allocation
- Theme exposures
- Factor exposures
- Geographic exposure
- Currency exposure
- Current theses
- Rebalance proposals
- Replacement candidates
- Expected income
- Scenario analysis
- Long-term performance

20.9 News, Events, and Sentiment Center

Show:

- Prioritized event feed
- Source confidence
- Entity links
- Affected assets
- Expected horizon
- Sentiment
- Narrative acceleration
- Contradictory reports
- Unverified rumors
- Price reaction
- Historical analogues

20.10 Execution Console

Show:

- Order proposals
- Risk decisions
- Open orders
- Partial fills
- Completed fills
- Rejections
- Broker messages
- Reconciliation status
- Execution quality
- Emergency controls

20.11 Data Health Center

Show:

- Provider status
- Missing data
- Stale data
- Divergent prices
- Delayed jobs
- Failed pipelines
- Dataset freshness
- Data-quality score
- Licensing status

20.12 Model Registry

Show:

- Model name
- Version
- Purpose
- Features
- Training range
- Validation range
- Test results
- Calibration
- Drift
- Approval status
- Deployment status
- Rollback version

20.13 Audit Center

Allow filtering by:

- User
 - Service
 - Strategy
 - Instrument
 - Order
 - Model
 - Date
 - Event type
 - Manual override
 - Risk rejection
 - Configuration change
-

21. NATURAL-LANGUAGE RESEARCH ASSISTANT

Create a private research assistant that can answer questions such as:

- Why is gold rising?
- Which current signals have the strongest out-of-sample support?
- Which positions are exposed to rising real yields?
- What changed in the portfolio risk today?
- Which strategies are underperforming their expected regime?
- What major events are scheduled this week?
- Which holdings have deteriorating fundamentals?
- Which strategies have recently experienced abnormal slippage?
- What happens to the portfolio under a 15% equity decline?
- Which investment themes have improving earnings revisions?
- Why was this order rejected?
- Which signal inputs are contradictory?

The assistant must:

- Cite internal data sources.
- State timestamps.
- Distinguish facts from inference.
- State uncertainty.
- Avoid unsupported causal claims.
- Never fabricate missing data.
- Never execute live trades through natural-language text alone.
- Require explicit structured approval for sensitive actions.

22. DATABASE DOMAIN MODEL

Create normalized and auditable schemas for at least:

```
users
roles
permissions
accounts
brokers
broker_accounts
venues
instruments
instrument_identifiers
trading_sessions
market_calendars
bars
trades
quotes
order_book_snapshots
```

funding_rates
open_interest
liquidations
macro_series
macro_releases
fundamental_facts
financial_statements
corporate_actions
news_articles
news_events
social_posts
social_metrics
entities
entity_links
features
feature_definitions
feature_values
datasets
dataset_versions
strategies
strategy_versions
strategy_parameters
research_experiments
backtest_runs
backtest_trades
backtest_metrics
models
model_versions
model_predictions
regimes
signals
signal_evidence
risk_policies
risk_limits
risk_decisions
portfolios
portfolio_allocations
positions
position_lots
orders
order_events
fills
cash_balances
performance_snapshots
reconciliation_runs
alerts
incidents
job_runs
data_quality_events
audit_logs

```
approvals
configuration_versions
```

High-volume time-series tables may live in analytical storage while metadata references remain in PostgreSQL.

Use immutable event records where appropriate.

23. SECURITY REQUIREMENTS

Implement:

- Strong authentication
- Multi-factor authentication where supported
- Least-privilege authorization
- Secret management
- Encryption in transit
- Encryption at rest
- Key rotation
- Session expiration
- Secure cookies
- CSRF protection
- Input validation
- Output encoding
- SQL-injection prevention
- SSRF prevention
- Rate limiting
- Audit logs
- IP restrictions as an optional control
- Dependency scanning
- Container scanning
- Static analysis
- Secret scanning
- Backup encryption
- Restore tests

Never store API secrets in:

- Source code
- Git history
- Frontend bundles
- Logs
- Error messages
- Screenshots
- Analytics systems

Broker credentials must be isolated from general application services.

Live-order submission should use a narrowly scoped execution service.

24. RELIABILITY AND OBSERVABILITY

Implement:

- Structured logging
- Metrics
- Distributed tracing
- Health endpoints
- Readiness checks
- Liveness checks
- Queue-depth monitoring
- Data-latency monitoring
- Job-duration monitoring
- Error-rate monitoring
- Broker-connectivity monitoring
- Order-lifecycle monitoring
- Reconciliation monitoring
- Model-drift monitoring
- Strategy-performance monitoring
- Cost monitoring

Define service-level objectives for:

- Data freshness
- Signal latency
- Order submission
- Reconciliation
- Dashboard availability
- Recovery time
- Recovery point

Create alerts with severity:

INFO
WARNING
HIGH
CRITICAL

Critical alerts include:

- Unknown broker positions
- Duplicate-order suspicion
- Reconciliation failure
- Global risk-limit breach
- Stale data used by active strategy
- Unauthorized live-trading activation
- Database corruption
- Kill-switch failure

- Secret exposure
 - Unexplained capital change
-

25. DISASTER RECOVERY

Document and test:

- Database backups
- Object-storage backups
- Configuration backups
- Model-artifact backups
- Audit-log retention
- Encryption-key recovery
- Broker-state recovery
- Order-state recovery
- Position reconciliation after outage
- Service restoration
- Regional outage response
- Provider failover
- Rollback
- Incident communication

Simulate:

- Database failure
 - Message-broker failure
 - Redis failure
 - Data-provider failure
 - Broker API failure
 - Partial deployment
 - Network interruption during order submission
 - Duplicate webhook or stream event
 - Clock drift
 - Corrupt market data
-

26. TESTING STANDARD

Implement:

- Unit tests
- Integration tests
- Contract tests
- Property-based tests
- Data-quality tests
- Backtest-regression tests
- Broker-adapter tests
- Risk-engine tests

- Security tests
- End-to-end tests
- Load tests
- Failure-injection tests
- Restore tests
- UI tests

Critical invariants must include:

- No order bypasses the Risk Engine.
- No disabled strategy creates a live order.
- No expired signal creates an order.
- No duplicate intent creates duplicate exposure.
- No live order is possible when live trading is disabled.
- No missing authorization permits account changes.
- No stale market data creates a valid signal.
- No unresolved reconciliation permits risk increases.
- No strategy modifies its own risk limits.
- No model deploys without approval.
- No backtest accesses future data.

Maintain golden datasets and deterministic regression cases.

27. CI/CD AND CHANGE MANAGEMENT

Every pull request must run:

- Formatting
- Linting
- Type checking
- Unit tests
- Integration tests
- Security scanning
- Migration validation
- Backtest-regression tests
- Build
- Container validation

Sensitive changes require additional review:

- Risk policy
- Position sizing
- Broker adapter
- Order submission
- Authentication
- Secrets
- Live-trading flags
- Database migrations
- Model promotion
- Kill switches

Deployments must support:

- Versioned artifacts
 - Rollback
 - Database migration safety
 - Feature flags
 - Canary or staged rollout
 - Paper environment before live environment
-

28. MULTI-AGENT DEVELOPMENT ORGANIZATION

Use specialized development agents with explicit responsibilities.

28.1 Chief Architect Agent

Responsible for:

- System boundaries
- Architecture decisions
- Dependency control
- Roadmap integrity
- Technical coherence

28.2 Quant Research Agent

Responsible for:

- Strategy hypotheses
- Statistical validation
- Backtesting methodology
- Multiple-testing controls
- Performance analysis

28.3 Data Engineering Agent

Responsible for:

- Data ingestion
- Normalization
- Point-in-time correctness
- Provenance
- Quality
- Storage

28.4 ML and NLP Agent

Responsible for:

- Feature pipelines
- Models
- News intelligence
- Sentiment
- Calibration
- Drift
- Model registry

28.5 Risk Officer Agent

Responsible for:

- Risk policies
- Limits
- Stress tests
- Kill switches
- Strategy approval criteria

The Risk Officer Agent may block implementation decisions that weaken safety.

28.6 Execution Engineer Agent

Responsible for:

- Broker adapters
- OMS
- EMS
- Idempotency
- Reconciliation
- Execution analysis

28.7 Backend Agent

Responsible for:

- APIs
- Domain services
- Authentication
- Database access
- Business logic

28.8 Frontend and Product Agent

Responsible for:

- Operator workflows

- Dashboard
- Charts
- Information architecture
- Accessibility
- Performance

28.9 DevOps and SRE Agent

Responsible for:

- Infrastructure
- CI/CD
- Monitoring
- Backup
- Recovery
- Cost controls

28.10 Security Agent

Responsible for:

- Threat modeling
- Secret protection
- Authorization
- Vulnerability review
- Auditability

28.11 QA and Red-Team Agent

Responsible for:

- Breaking assumptions
- Failure testing
- Backtest-bias detection
- Order-safety testing
- Adversarial input testing

Agents must not independently modify the same high-risk files without coordination.

Architecture decisions must be documented in `DECISIONS.md`.

29. PHASED IMPLEMENTATION ROADMAP

Phase 0 — Discovery and Architecture

Deliver:

- Repository audit

- Requirements
- Architecture
- Provider matrix
- Broker matrix
- Security model
- Risk policy
- Roadmap
- Cost analysis

Exit criteria:

- Scope is documented.
- Architecture is reviewed.
- Data licenses are understood.
- No live execution exists.

Phase 1 — Platform Foundation

Deliver:

- Monorepo or organized repository
- Authentication
- Core database
- Instrument master
- Configuration system
- Audit logging
- Local infrastructure
- CI/CD
- Observability foundation

Phase 2 — Historical Data Platform

Deliver:

- Market-data ingestion
- Macro ingestion
- Corporate actions
- Data normalization
- Data-quality checks
- Versioned datasets
- Point-in-time data contracts

Phase 3 — Quant Research Foundation

Deliver:

- Feature engine
- Strategy interface
- Vectorized backtesting
- Cost models
- Experiment tracking

- Performance reports
- Bias tests

Phase 4 — Event-Driven Backtesting

Deliver:

- Order simulation
- Partial fills
- Broker constraints
- Funding
- Borrow
- Corporate actions
- Reconciliation simulation

Phase 5 — Initial Strategy Library

Implement transparent baseline strategies:

- Trend
- Mean reversion
- Breakout
- Momentum
- Factor
- Macro
- Crypto funding or basis where feasible

Do not implement every strategy at once.

Phase 6 — Risk Engine

Deliver:

- Position limits
- Portfolio limits
- Pre-trade checks
- Drawdown controls
- Kill switches
- Stress testing
- Risk dashboard

Phase 7 — Operator Dashboard

Deliver:

- Command center
- Market intelligence
- Instrument workstation
- Strategy lab
- Backtest viewer

- Risk center
- Audit center

Phase 8 — News and Sentiment Intelligence

Deliver:

- News ingestion
- Entity linking
- Event extraction
- Sentiment
- Narrative clusters
- Source scoring
- Event timeline

Phase 9 — Regime and Ensemble Engine

Deliver:

- Regime probabilities
- Strategy-regime evaluation
- Strategy ensemble
- Capital allocation
- Correlation-aware risk budgets

Phase 10 — Investment Portfolio System

Deliver:

- Theme engine
- Fundamental scoring
- Portfolio construction
- Rebalancing
- Thesis records
- Scenario analysis

Phase 11 — Broker Paper Trading

Deliver:

- Broker adapter
- Paper orders
- Order lifecycle
- Reconciliation
- Execution quality
- Alerts

Phase 12 — Shadow Mode

Deliver:

- Live-data signals
- Hypothetical orders
- Realistic fill comparison
- Backtest-to-live divergence reports
- Operational incident testing

Phase 13 — Limited Live Pilot

Only after explicit approval.

Deliver:

- Minimal capital allocation
- One strategy
- One broker
- One asset class
- Strict limits
- Manual kill switch
- Daily reconciliation
- Daily risk report

Phase 14 — Controlled Expansion

Expand only when evidence supports it:

- Additional strategies
- Additional instruments
- Additional asset classes
- Additional brokers
- Higher capital within approved limits

30. MVP SCOPE CONTROL

The final architecture may support thousands of instruments, but the initial verified MVP must not attempt to trade 1,000–2,000 instruments simultaneously.

Begin with a controlled universe such as:

- Highly liquid United States equities or ETFs
- Major forex pairs
- Gold
- Bitcoin and Ethereum
- A limited set of highly liquid crypto instruments

Prove:

- Data quality
- Backtest correctness
- Signal stability
- Risk controls
- Broker reliability
- Reconciliation
- Operational monitoring

Then expand the universe.

Universe expansion must be based on:

- Data availability
- Liquidity
- Execution quality
- Strategy capacity
- Broker support
- Cost
- Legal availability
- Risk

31. COST GOVERNANCE

Track operational costs by:

- Data provider
- News provider
- Social provider
- Cloud compute
- Storage
- Database
- Streaming
- AI inference
- Broker fees
- Exchange fees
- Monitoring
- Backup

Create budget modes:

```
LOCAL_RESEARCH  
LOW_COST_PAPER  
PROFESSIONAL_PAPER  
LIMITED_LIVE  
SCALED_LIVE
```

The system must estimate whether an additional dataset or model creates enough measurable value to justify its cost.

Do not use expensive AI inference for calculations that deterministic code can perform.

32. REQUIRED REPORTS

Generate:

- Daily market report
- Daily risk report
- Daily execution report
- Daily data-health report
- Weekly strategy report
- Weekly model-drift report
- Weekly portfolio report
- Monthly investment review
- Monthly strategy attribution
- Monthly cost report
- Monthly incident report
- Quarterly model-governance review
- Live-readiness report

Every report must distinguish:

- Facts
 - Model estimates
 - Inferences
 - Unverified information
 - Missing data
-

33. FINAL ACCEPTANCE CRITERIA

The system is not considered complete merely because the interface loads or signals appear.

A professional release requires:

- Reproducible datasets
- Reproducible backtests
- Point-in-time correctness
- Realistic costs
- Bias controls
- Strategy versioning
- Model versioning
- Configuration versioning
- Independent Risk Engine
- Tested kill switches

- Broker reconciliation
 - Full audit trail
 - Data-health monitoring
 - Incident monitoring
 - Backups
 - Restore verification
 - Security review
 - Paper-trading evidence
 - Shadow-mode evidence
 - Documented limitations
 - Explicit live-trading approval
-

34. DEVELOPMENT EXECUTION RULES

Follow this workflow for every implementation cycle:

1. Inspect the relevant code and documentation.
2. State the exact objective internally.
3. Identify dependencies and risks.
4. Update the roadmap.
5. Implement the smallest complete vertical slice.
6. Add tests.
7. Run tests.
8. Validate data and security impacts.
9. Update documentation.
10. Record evidence.
11. Update current status.
12. Do not silently defer failures.

For each completed cycle report:

```
Cycle objective:
Files changed:
Architecture impact:
Data impact:
Security impact:
Risk impact:
Tests added:
Tests executed:
Results:
Known limitations:
Roadmap items completed:
Roadmap items remaining:
Next highest-priority task:
```

Do not produce fake test results.

Do not state that a feature works unless it has been executed and verified.

35. FIRST EXECUTION ASSIGNMENT

Begin now with Phase 0.

Perform the following actions:

1. Audit the complete repository.
2. Map all existing components.
3. Identify the current technology stack.
4. Identify any existing trading, data, AI, dashboard, broker, or portfolio components.
5. Identify security weaknesses.
6. Identify architecture weaknesses.
7. Identify missing tests.
8. Identify data-provider requirements.
9. Identify broker requirements.
10. Identify legal, licensing, and operational constraints.
11. Create all required documentation skeletons.
12. Produce the first complete version of `MASTER_ROADMAP.md`.
13. Produce `SYSTEM_ARCHITECTURE.md`.
14. Produce `DATA_ARCHITECTURE.md`.
15. Produce `RISK_MANAGEMENT_POLICY.md`.
16. Produce `BACKTESTING_STANDARD.md`.
17. Produce a prioritized Phase 1 implementation plan.
18. Do not activate live trading.
19. Do not add real broker credentials.
20. Do not place orders.
21. Do not skip directly to advanced AI models.
22. Do not start broad implementation until the foundation is documented.

The first response after the audit must include:

- Current repository state
- Reusable components
- Critical risks
- Proposed architecture
- Proposed initial market universe
- Proposed initial data sources
- Proposed initial broker integration
- Phase-by-phase roadmap
- Cost-sensitive MVP plan
- Major design decisions
- Blocking questions that cannot be resolved from the repository
- Exact files created or updated

Proceed autonomously within these constraints. Prioritize correctness, capital preservation, auditability, reproducibility, and realistic economic performance over speed, visual effects, trade frequency, or apparent sophistication.